
OpenFedNow

Bridging Legacy Core Banking to Both U.S. Instant Payment Rails:
FedNow and RTP

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This document explains, in plain language, what problem this framework addresses, why that problem matters for the United States, and what specific contribution the author has made — drawing on demonstrated experience from Brazil's national payment system.

Section 1 — The Problem: Why Most U.S. Banks Face Barriers to Instant Payment Participation

The United States Has Two Instant Payment Rails

As of 2026, the United States operates two ISO 20022 instant payment networks. The Federal Reserve's FedNow Service, launched in July 2023, and The Clearing House's RTP® network, launched in 2017, together form the infrastructure for real-time payments in the U.S. Both are designed to support near-real-time account-to-account payments, available any hour, any day of the year.

	FedNow Service	RTP® Network
Operator	Federal Reserve (public)	The Clearing House (private)
Launched	July 2023	November 2017
Participants	~1,700 institutions	~1,100 institutions
Transaction limit	Up to \$1M (rising to \$10M)	Up to \$10M
Settlement	Fed Master Account	TCH prefunded joint account
Message standard	ISO 20022 (JSON envelope)	ISO 20022 (XML envelope)
Availability	24/7/365	24/7/365

Industry data confirms that dual-rail participation is increasingly emerging as a practical strategy among instant-payment-enabled institutions. Among institutions currently enabled for instant payments, 58% participate in both FedNow and RTP. Jack Henry — one of the three dominant U.S. core banking vendors — formally recommends that financial institutions "receive on both networks" as the baseline strategy. The question for most institutions is no longer *which rail* to choose, but *how* to connect to either one.

The Adoption Gap — On Both Rails

Despite two operating networks and broad institutional intent, the vast majority of U.S. financial institutions still face significant barriers to real-time payment participation. Of the nation's 10,000+ eligible banks, credit unions, and depository institutions, fewer than 1,700 are connected to FedNow and fewer than 1,100 to RTP — and the overlap is significant. Most institutions are not fully connected to either network, let alone both.

10,000+	Financial institutions eligible to participate in FedNow and RTP
~1,700	Institutions connected to FedNow (as of early 2026)
~1,100	Institutions connected to RTP (as of early 2026)
73%	Of non-connected institutions citing legacy core systems as the primary barrier (U.S. Faster Payments Council / Finzly, 2024)

58%

Of instant-payment-enabled institutions that use both rails — dual-rail adoption is increasingly common among enabled institutions

The reason most banks have not connected is not simply lack of interest; the central technical barrier is legacy core integration, which also drives cost and implementation complexity. According to the U.S. Faster Payments Council, 73 percent of financial institutions identify their existing core banking systems as a moderate-to-severe obstacle to instant payment participation. This barrier applies to both FedNow and RTP in materially similar ways — because the same core banking vendors serve the same institutions on both networks.

What Is a Core Banking System?

Every bank runs on software called a "core banking system" — the central platform that manages accounts, tracks balances, and processes transactions. Three technology vendors supply these systems to the vast majority of U.S. banks:

- Fiserv supplies core systems to approximately 42% of U.S. banks
- Jack Henry supplies core systems to approximately 21% of U.S. banks
- FIS supplies core systems to approximately 9% of U.S. banks

Together, these three vendors serve more than 70 percent of the U.S. banking market. These systems were built decades ago for batch processing — not for the 24/7 real-time requirements of either FedNow or RTP.

Why Legacy Systems Cannot Connect to Either Rail

The four incompatibilities that block FedNow participation are materially similar to those that block RTP participation. Both networks impose real-time requirements on institutions whose core systems were designed for a different era.

Both FedNow and RTP require:	Legacy Core Banking cannot:
Each transaction processed individually in under 20 seconds (FedNow) / 10 seconds (RTP)	Processes transactions in batches — typically overnight
24/7/365 availability — no maintenance windows allowed	Goes offline periodically for maintenance and updates
ISO 20022 international messaging standard	Uses proprietary, vendor-specific data formats and protocols
High volumes of concurrent real-time transactions	Was not designed for high concurrent transaction volumes

These are not minor technical adjustments. They represent fundamentally different design philosophies — one built for the banking world of the 1970s and 1980s, the other for a 24/7 digital economy. Replacing the core banking system itself is not a realistic option: these systems are deeply integrated into bank

operations, cost tens of millions of dollars to replace, and take five to ten years to migrate.

The result: most U.S. banks face significant barriers to participation in either instant payment network — not because they lack interest, but because no accessible, affordable, vendor-neutral technical solution has been available to bridge their existing systems to the new infrastructure.

Section 2 — Why Existing Approaches Do Not Solve the Problem

Approach 1: Point-to-Point Custom Integration

Some banks have hired technology firms to build a custom bridge to FedNow or RTP. This can work — but it is expensive, slow, not reusable, and applies to only one rail at a time. Each bank pays \$500,000 to \$2 million for work that solves the same problem every other bank faces. And if the institution later wants to add the second rail, the integration work begins again from scratch.

Approach 2: Waiting for the Core Vendor

In theory, Fiserv, Jack Henry, and FIS could update their platforms to natively support both rails. In practice, these vendors have large customer bases, long development cycles, and limited incentive to accelerate. Many community banks have been waiting for their core vendor to provide a FedNow integration path for years. Even where vendor solutions exist, they are typically proprietary, limited to that vendor's customers, and do not provide a path to both rails simultaneously.

Approach 3: Core System Replacement

Newer cloud-native platforms are designed for real-time payments. But migrating a bank's core system takes three to seven years and costs tens of millions of dollars, with substantial operational risk. For the community banks and credit unions that make up the majority of unconnected institutions, this is not a practical solution.

The Gap That Remains

The record does not identify a broadly accessible, vendor-neutral, open-source middleware solution that allows a bank to connect its existing core system to both FedNow and RTP without replacing it. The central insight behind OpenFedNow is that **the same core processing layers work regardless of which core banking vendor the institution uses — and regardless of which payment rail delivers the message.** A current source-line analysis shows that the majority of the implementation consists of shared gateway and framework components, while the remaining code is concentrated in the vendor adapter packages. The important architectural point is that Layers 2–4 remain shared across rails and vendors, while only Layer 1 varies by rail and the adapter layer varies by core banking platform.

This means a single shared framework, built once and completed through bounded vendor-specific adapters, could serve thousands of institutions — on both rails — across all three major core banking platforms.

Section 3 — The Proven Approach: Brazil's Instant Payment System (PIX)

What Is PIX?

PIX is Brazil's national instant payment system, launched November 16, 2020, under a mandate from the Banco Central do Brasil. Within its first year, PIX became the most commonly used payment method in Brazil. On its peak single day, the national PIX network processed 313.3 million transactions. Today, PIX has over 175 million registered users — more than 80% of Brazil's adult population.

Nov. 16, 2020	PIX launch date under Central Bank of Brazil mandate
175M+	Registered users — more than 80% of Brazil's adult population
313.3M	Single-day transaction record across the national PIX network
#1	Most-used payment method in Brazil within its first year of operation

The Same Problem at National Scale

When PIX launched, Brazilian banks faced the same class of legacy-core integration challenge confronting U.S. institutions: existing core banking systems — many running on IBM mainframe computers — were built for batch processing and could not natively connect to the new real-time network.

Santander Brazil, one of the country's largest banks, faced this challenge directly. The author, Daniel Stelzer Mori, served as Application Architect and Technical Anchor for the integration at Santander Brazil, serving in an architectural role for the integration between the bank's legacy IBM z/OS mainframe systems and the PIX network. This was a production system that processed real financial transactions for millions of customers from PIX's launch day.

What the Author Built

The solution the author helped design and implement for PIX at Santander Brazil is a five-layer software architecture in which each layer solves one of the four incompatibilities. Two components were novel design contributions:

- **The Anti-Corruption Layer:** Translates between ISO 20022 and the proprietary internal language of each core banking vendor — without modifying the core system.
- **The Shadow Ledger:** Maintains real-time balance tracking so a bank can continue processing payments even when its core system is offline for maintenance — resolving the 24/7 availability requirement.

The methodology underlying these components was applied in a live production environment at one of Brazil's largest banks, processing millions of transactions during the busiest payment periods in the country's history.

Why PIX, FedNow, and RTP Address the Same Core Challenge

PIX, FedNow, and RTP all use the same international messaging standard — ISO 20022. All three networks impose the same four real-time requirements on legacy core banking systems. The five-layer

architecture the author applied at Santander Brazil resolves all four incompatibilities in a way that is structurally transferable to the U.S. context. OpenFedNow reflects a production-informed methodology based on that experience — adapted for U.S.-specific core banking vendors and the regulatory environment of the Federal Reserve and The Clearing House.

Section 4 — OpenFedNow: Dual-Rail Architecture and Current State

What Is OpenFedNow?

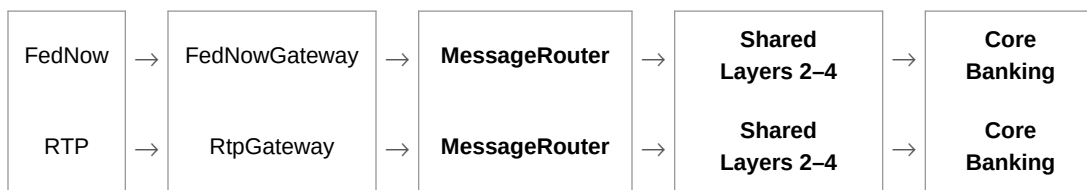
OpenFedNow is an open-source middleware framework that enables any U.S. financial institution to connect its existing core banking system to both the Federal Reserve's FedNow Service and The Clearing House's RTP® network — without replacing or modifying its existing core system. It is released under the Apache 2.0 license, making it free to use for any bank, credit union, or technology vendor.

The framework is publicly available at github.com/danielismori/open-fednow.

Rail-Agnostic Architecture — The Core Design Insight

The most important architectural property of OpenFedNow is that Layers 2 through 4 — the Anti-Corruption Layer, the Processing Engine, and the Shadow Ledger — have no knowledge of which payment rail delivered a message. They operate on parsed ISO 20022 message objects. This is not an aspiration; it is a deliberate design decision documented in Architecture Decision Record ADR-0005, and it is reflected in the codebase.

The gateway layer (Layer 1) is the only variation point between rails:



Both gateways parse their respective message envelopes and deliver the same internal message object to the same processing pipeline. An institution deploying OpenFedNow for FedNow today does not need to re-integrate Layers 2–4 when adding RTP participation — only the remaining Layer 1 RTP live-connectivity components need to be completed. This reduces the marginal cost of adding the second rail dramatically.

How It Resolves Each Incompatibility

The framework addresses all four incompatibilities for both FedNow and RTP simultaneously:

- **Speed mismatch:** The SyncAsyncBridge manages the handoff between FedNow's 20-second and RTP's real-time requirements and the legacy system's need for more processing time.
- **Availability mismatch:** The Shadow Ledger maintains real-time balance tracking independently of the core system. Payments arriving during core maintenance are queued, answered with a provisional acceptance, and reconciled when the core returns — for both rails simultaneously.
- **Protocol mismatch:** The Anti-Corruption Layer translates between the ISO 20022 canonical model and each core banking vendor's proprietary format. Rail-specific translation (FedNow JSON vs. RTP XML) is handled in Layer 1 only.
- **Concurrency mismatch:** Saga orchestration and idempotency key management reduce duplicate execution risk and provide exactly-once processing behavior through idempotency controls — regardless of which rail the payment arrived on.

Current Development Status

As of May 2026, the core framework — Layers 2 through 4, the Shadow Ledger (with Redis-backed optimistic locking and 24/7 bridge mode), the JDBC-backed Saga orchestration engine, the idempotency service, and the reconciliation service — is implemented and tested. The repository includes integration tests running against real PostgreSQL, Redis, and RabbitMQ infrastructure using Testcontainers. The FedNow gateway (Layer 1), including the inbound and outbound payment flows (sandbox/reference mode), is implemented. Admin endpoint security is implemented via HTTP Basic authentication with BCrypt.

The RTP gateway (Layer 1) is implemented and tested in reference mode. It includes canonical ISO 20022 XML parsing with XXE protection, outbound RTP XML serialization, RTP gateway routing, HTTP sandbox client behavior, and reference-mode send/receive flows. Remaining live-production items — including The Clearing House certificate validation, private-network transport, institutional onboarding, and formal RTP certification — are external production dependencies, not unresolved framework design gaps.

All three vendor adapters — Fiserv, FIS, and Jack Henry — are fully implemented and tested. Each implements the CoreBankingAdapter interface with OAuth 2.0 token management, vendor-specific ISO 20022 reason-code mapping, and WireMock-based integration tests covering the core payment scenarios. The SandboxAdapter covers all payment scenarios, and a MockVendorAdapter provides a reference implementation with a configurable in-memory balance ledger for contract testing.

Who Can Use It

OpenFedNow is designed for any U.S. financial institution running Fiserv, Jack Henry, or FIS core banking platforms, including:

- Community banks — limited technology staff, unable to fund expensive custom integration for each rail separately
- Credit unions — member-owned cooperatives currently excluded from both rails
- Mid-sized regional banks seeking a faster, lower-cost path to dual-rail participation

Section 5 — Transferability: From Brazil to the United States

Technical Alignment

PIX, FedNow, and RTP all use ISO 20022 as their messaging standard. The data formats, message types, and processing logic developed for PIX are the same ones used by both U.S. rails. The author's production experience with ISO 20022 at national scale is applicable to the U.S. context — for both networks simultaneously.

Structural Alignment

The architectural challenge — legacy batch-processing core banking systems that cannot natively support a real-time national payment network — represents the same class of legacy-core integration challenge across PIX, FedNow, and RTP, though each rail has its own network, certification, message-envelope, and regulatory requirements. The five-layer architecture addresses that shared challenge and transfers across all three networks.

The Shared-Framework Principle — Compounded

A current source-line analysis of the OpenFedNow implementation shows that the majority of the codebase consists of shared gateway and framework components, while the remaining code is concentrated in the vendor adapter packages. OpenFedNow extends this principle across rails: Layers 2 through 4 — the Anti-Corruption Layer core, the Processing Engine, and the Shadow Ledger — are also rail-agnostic, because they operate on internal domain objects with no knowledge of which payment rail delivered a message. Only Layer 1 varies per rail.

This means: build once, serve both FedNow and RTP, across all three major U.S. core banking platforms. The marginal cost of adding the second rail, once the first is integrated, is confined to Layer 1 only.

What Is Genuinely New for the U.S. Context

U.S. core banking vendors use different proprietary protocols than Brazilian mainframe systems. Federal Reserve PKI and TCH certificate requirements are U.S.-specific. The vendor adapters for Fiserv, Jack Henry, and FIS are new work that did not exist for the PIX project. OpenFedNow is an original implementation applying proven methodology to the specific requirements of U.S. core banking vendors and both U.S. instant payment rails.

Section 6 — Expected Impact on the United States

Expanding Access to Both Instant Payment Networks

If OpenFedNow is successfully completed and adopted, it would provide a ready-made dual-rail integration path for the thousands of U.S. financial institutions currently facing barriers to participation in FedNow, RTP, or both. Because the framework is open-source and available at no cost, it could reduce a major barrier for community banks and credit unions by lowering the custom-development burden across two networks.

A community bank with 10,000 customers and a regional credit union with 50,000 members would have access to the same dual-rail integration infrastructure as a large national bank — because the hard architectural work has been done once and made freely available.

Benefits to Consumers and Businesses

Broader instant payment participation has direct, tangible benefits for bank customers:

- Payroll disbursements received immediately, reducing overdraft risk for workers paid biweekly
- Small business invoices paid and confirmed instantly, improving cash flow
- Government benefit payments — including emergency relief and disaster assistance — reaching recipients in seconds
- Peer-to-peer transfers without waiting for business-day settlement
- Real-time payment confirmations reducing fraud and disputes

Strengthening the Competitiveness of U.S. Payment Infrastructure

The United States is unusual in having two parallel instant payment networks. Brazil's PIX achieved near-universal institutional adoption through a single mandated network. The U.S. dual-rail model requires that institutions integrate with two independent systems — doubling the integration burden without open-source middleware to share the work. OpenFedNow directly addresses this by providing a single shared integration framework that could support dual-rail participation for institutions of different sizes, subject to vendor-specific implementation, institutional credentials, and rail certification.

Alignment with Federal Reserve and Industry Policy Goals

The Federal Reserve has stated that broad participation across institutions of all sizes is a primary goal for FedNow. Jack Henry — one of the three target vendors — formally recommends that institutions receive on both networks. The U.S. Faster Payments Council has identified legacy core integration as the primary barrier. OpenFedNow directly advances all three policy objectives by providing the missing piece: a free, open-source, vendor-neutral, dual-rail integration framework that could support dual-rail participation for institutions of different sizes, subject to vendor-specific implementation and certification.

~72%

Core platforms serving approximately 72% of U.S. banks are addressed by the implemented Fiserv, FIS, and Jack Henry reference-mode adapters, with substantial additional credit-union coverage

~8,300+	FedNow-eligible institutions not yet connected, with RTP participation also substantially below full market coverage
\$0	Licensing cost to any institution adopting the open-source framework (Apache 2.0)
2 rails	Covered by a single integration — FedNow and RTP — through the rail-agnostic architecture

Conclusion

The United States has built two instant payment networks — the Federal Reserve's FedNow Service and The Clearing House's RTP network. Both are operational, well-funded, and endorsed by industry and federal policy. Both use the same ISO 20022 messaging standard. Yet most U.S. financial institutions still face significant barriers to participation in either — blocked by the same legacy core banking barrier on both rails.

The author, Daniel Stelzer Mori, has direct, production-validated experience designing the same class of legacy-to-instant-payment integration solution. The five-layer middleware architecture he helped design and implement at Santander Brazil enabled that bank to connect its legacy mainframe systems to PIX from launch day — a system that now serves more than 175 million users and processes hundreds of millions of transactions daily.

OpenFedNow applies that proven architecture to the U.S. context — with a critical extension: the same integration covers both FedNow and RTP simultaneously, because Layers 2 through 4 are rail-agnostic by design. A community bank that deploys OpenFedNow could pursue dual-rail connectivity through a single shared integration framework, subject to vendor-specific implementation, institutional credentials, and rail certification.

The work addresses a specific, documented gap in U.S. financial infrastructure — one that is larger than any single network. It draws on rare production experience from a national-scale instant-payment implementation and applies that experience to a documented U.S. infrastructure gap. The core framework and all three vendor adapter implementations are complete and publicly available — the remaining step is institutional onboarding and certification, not missing architectural work, and the path to dual-rail access for thousands of institutions is now an implementation decision, not a research problem.

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